

# SatsPath × Arkade Wallet — MVP Technical Integration Plan

**Objective:** Enable Arkade wallet users to send/receive Bitcoin via human-readable aliases (e.g., Truja@sexo.ya) with automatic rail selection (Lightning → On-chain → Ark), client-side VT XO verification, and encrypted identity management.

**Timeline:** 4-6 weeks | **Approach:** WASM bindings from Rust core to TypeScript wallet

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## Current Architecture Summary

### SatsPath (Rust) — Existing Components

| Crate           | Purpose                                             | Key Exports                                                    |
|-----------------|-----------------------------------------------------|----------------------------------------------------------------|
| satspath-core   | Types, crypto, resolvers, registry, profile signing | PaymentMethod, SignedPaymentProfile, ChainResolver, Registry   |
| satspath-router | Rail selection logic                                | quote(), select_route(), RouteQuote, FeeEstimate               |
| satspath-cli    | CLI: init, register, quote, pay, preview            | —                                                              |
| satspathd       | HTTP daemon (/v1/quote, /v1/pay, /v1/resolve)       | —                                                              |
| satspath-wasm   | WASM bindings (crypto only today)                   | verify_signed_profile, canonical_profile_json, topic_for_alias |

### Arkade SDK (TypeScript) — Existing Components

- **Tier 1:** VT XO DAG reconstruction + Schnorr/MuSig2 validation
- **Tier 2:** Taproot tree audit + CSV delays + Boltz HTLC support
- **Tier 3:** Sovereign exit — local encrypted storage + ASP-independent broadcast
- **Interfaces:** IndexerProvider, OnchainProvider, StorageProvider

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## Integration Strategy: WASM Bindings

Extend satspath-wasm to expose **resolver chain + router** to TypeScript. Arkade wallet calls:

```
const quote = await satspath.quote("Truja@sexo.ya", 21000n);  
// → { selected_method, reason, fee_sats, eta, qr, execution_mode }
```

**Why WASM:** Single binary, works in browser/PWA/React Native, no sidecar process.

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## MVP Scope (4-6 Weeks)

### Phase 1: Extend satspath-wasm (Week 1-2)

**Goal:** Expose resolver + router to TypeScript via WASM

**Tasks:** 1. Add wasm feature flags to satspath-core and satspath-router (remove tokio/request dependencies) 2. Create WASM-compatible resolver implementations using web-sys fetch 3. Export quote(), verify\_signed\_profile(), ChainResolver, Registry from satspath-wasm 4. Add fingerprint\_pubkey(), mask\_identifier() helpers

**Files to modify:** - crates/satspath-core/Cargo.toml — add wasm feature - crates/satspath-router/Cargo.toml — add wasm feature - crates/satspath-wasm/src/lib.rs — new exports - crates/satspath-wasm/src/resolver.rs — NEW: WASM resolver chain - crates/satspath-wasm/src/router.rs — NEW: router wrapper

## Phase 2: Build WASM Package (Week 2)

**Goal:** Publish @satspath/wasm npm package

**Tasks:** 1. Configure wasm-pack build for web target 2. Generate TypeScript definitions via wasm-bindgen 2. Publish to local npm registry or GitHub Packages 3. Verify import in TypeScript: import { quote } from '@satspath/wasm'

## Phase 3: Arkade Wallet Service Layer (Week 2-3)

**Goal:** Integrate WASM into Arkade wallet services

**New file:** arkade-wallet/src/services/satspath.ts

```
import { quote, verify_signed_profile, ChainResolver } from '@satspath/wasm';

export async function getPaymentQuote(alias: string, amountSats: bigint): Promise<PaymentQuote>
export function verifyProfile(profile: SignedPaymentProfile): boolean
export async function resolveProfile(alias: string): Promise<SignedPaymentProfile>
```

**Types:** arkade-wallet/src/types/satspath.ts (mirror Rust types)

## Phase 4: Send/Receive UI Flows (Week 3-4)

**Goal:** User-facing flows in Arkade wallet

**Send Flow** (SendFlow.tsx):

Input: alias + amount → satspath.quote() → shows rail + fee + QR → Pay button

**Receive Flow** (ReceiveFlow.tsx):

Manage identity keypair → Add payment methods → Sign profile → Publish (HTTPS/Nostr)

**Components:** - RailSelector.tsx — Shows selected rail, reason, fee, ETA - OwnershipBadges.tsx — / per method verification status - ArkVtxoStatus.tsx — VTXO verification progress + sovereign exit status

## Phase 5: Ark VTXO Verification (Week 4)

**Goal:** When Ark rail selected, run Arkade SDK verification

```
// Ark Receive
async function onArkReceive(vtxoOutpoint: Outpoint, arkMethod: ArkMethod) {
  const report = await verifyVtxoChain(vtxoOutpoint, indexer, onchain);
  if (report.valid) {
    await storage.setItem(`ark_exit:${vtxoOutpoint}`, encrypt(report.exitData));
    showUI("VTXO verified - sovereign exit ready");
  }
}

// Ark Send
async function sendViaArk(quote: RouteQuote, amountSats: bigint) {
  const intent = await arkade.createTransferIntent({
    server: arkMethod.server,
    recipientPubkey: arkMethod.pubkey,
    amountSats,
```

```
});  
}
```

## Phase 6: Profile Management + Ownership Proofs (Week 4-5)

Goal: Full identity lifecycle

- Generate secp256k1 identity keypair (encrypted at rest via Web Crypto API)
- Per-method ownership proofs:
  - **Lightning**: Sign LNURL-pay callback challenge
  - **On-chain**: Sign message with address key
  - **Ark**: ArkOwnershipProof from ASP
- Profile publication: HTTPS well-known → Nostr NIP-05

## Phase 7: Testing + E2E (Week 5-6)

- Testnet Lightning payment (LDK)
- Testnet On-chain payment (BDK)
- Testnet Ark payment + VTXO verification
- Fee estimation accuracy
- Error handling / edge cases

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## Technical Details

### WASM Feature Flags

```
# satspath-core/Cargo.toml  
[features]  
default = ["std"]  
wasm = [] # No tokio, request, tokio-tungstenite  
  
# satspath-router/Cargo.toml  
[features]  
default = ["std"]  
wasm = ["satspath-core/wasm", "satspath-router/fees-wasm"]  
  
# satspath-wasm/Cargo.toml  
[dependencies]  
satspath-core = { path = "../satspath-core", features = ["wasm"] }  
satspath-router = { path = "../satspath-router", features = ["wasm"] }  
wasm-bindgen = "0.2"  
wasm-bindgen-futures = "0.4"  
web-sys = { version = "0.3", features = ["Fetch", "Request", "Response", "Headers", "Window", "Crypto"] }
```

### WASM Resolver Chain (browser-compatible)

```
// satspath-wasm/src/resolver.rs  
use web_sys::{Request, RequestInit, RequestMode, Response};  
use wasm_bindgen_futures::JsFuture;  
  
pub struct WasmChainResolver {  
    local_registry: WasmRegistry,  
    bip353_resolver: WasmBip353Resolver,  
    https_resolver: WasmHttpsResolver,  
}
```

```

    nostr_resolver: WasmNostrResolver,
}

impl WasmChainResolver {
    pub async fn resolve_alias(&self, alias: &str) -> Result<SignedPaymentProfile, JsValue> {
        // Try each resolver in order: local + BIP353 + HTTPS + Nostr
    }
}

```

## TypeScript Types (auto-generated + manual)

```

// arkade-wallet/src/types/satspath.ts
export type PaymentMethod =
    | { type: "Lightning"; label: string; lightning_address?: string; lnurl?: string; bolt12?: string; re
    | { type: "Onchain"; label: string; network: "mainnet" | "testnet" | "regtest"; address?: string; sil
    | { type: "Ark"; label: string; server: string; pubkey: string; vtxo_pointer?: string; opaque_uri?: s

export interface PaymentProfile {
    alias: string;
    identity_pubkey: string;
    methods: PaymentMethod[];
    updated_at: number;
    expires_at?: number;
    sequence?: number;
    preferences: string[];
    nonce?: string;
    method_verifications: MethodVerification[];
}

export interface SignedPaymentProfile {
    profile: PaymentProfile;
    signature: string; // hex Schnorr
}

export interface RouteQuote {
    selected_method: PaymentMethod;
    reason: string;
    estimated_fee_sats: number;
    estimated_confirmation: string;
    fee_snapshot?: FeeSnapshot;
    execution_mode: "Preview" | "MainnetPreview" | "TestnetExperimental" | "ManualWallet";
    wallet_hint: string;
}

export interface PaymentQuote {
    rail: "Lightning" | "Onchain" | "Ark";
    rail_label: string;
    fee_sats: number;
    eta: string;
    reason: string;
    qr_payload: string; // BOLT11 / BIP21 / ark: URI
    ownership_verified: boolean;
    execution_mode: RouteQuote["execution_mode"];
}

```

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## File Structure (Post-Integration)

```
satspath/  
  crates/  
    satspath-core/  
      Cargo.toml          # +wasm feature  
    satspath-router/  
      Cargo.toml          # +wasm feature  
    satspath-wasm/  
      Cargo.toml  
      src/  
        lib.rs            # exports  
        crypto.rs         # existing  
        resolver.rs       # NEW: WASM resolver chain  
        router.rs         # NEW: router wrapper  
        helpers.rs        # NEW: fingerprint, mask  
      wasm-pack build --target web  
  docs/  
    TECHNICAL_PLAN_MVP.md # this file  
  sdk/wasm/pkg/           # generated npm package  
  
arkade-wallet/  
  src/  
    services/  
      satspath.ts         # NEW: SatsPath integration  
      arkade.ts           # existing Arkade SDK wrapper  
      lightning.ts        # LDK wrapper  
      onchain.ts          # BDK wrapper  
    components/  
      SendFlow.tsx  
      ReceiveFlow.tsx  
      RailSelector.tsx  
      OwnershipBadges.tsx  
      ArkVtxoStatus.tsx  
    types/  
      satspath.ts         # NEW: TS types mirror  
  package.json            # + @satspath/wasm dep
```

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## Risk Mitigation

| Risk                                       | Mitigation                                                                 |
|--------------------------------------------|----------------------------------------------------------------------------|
| WASM bundle too large                      | <code>wasm-opt -Oz</code> , tree-shake unused crate code, lazy-load        |
| Resolver chain fails in browser (CORS/DNS) | Fallback: HTTP daemon proxy for DNS/Nostr; HTTPS well-known works natively |
| Ark VTXO verification slow                 | Run async with progress UI; cache verified VTXOs                           |
| Profile publication complex                | MVP: HTTPS well-known only; Nostr/P2P post-MVP                             |
| Fee estimation inaccurate                  | Use mempool.space API + 10% buffer (already in router)                     |

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## Success Criteria (MVP Done)

- ☐ User enters `alias@domain` + amount → sees rail + fee + QR in < 2s
  - ☐ Lightning payment works via LDK (testnet)
  - ☐ On-chain payment works via BDK (testnet)
  - ☐ Ark payment shows VTXO verification status (testnet)
  - ☐ Profile creation + signing + HTTPS publish works
  - ☐ Ownership badges show / per method
  - ☐ All crypto in WASM (no native deps)
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## Next Steps

1. **Approve plan** → proceed to implementation
  2. **Phase 1:** Add `wasm` feature flags to `satspath-core` and `satspath-router`
  3. **Phase 1:** Implement `WasmChainResolver` in `satspath-wasm`
  4. **Phase 1:** Export `quote()` and router types from `satspath-wasm`
  5. **Phase 2:** `wasm-pack build` → verify TypeScript imports
  6. **Phase 3:** Build Arkade wallet `satspath.ts` service
  7. **Phase 4:** Wire UI components
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*Generated as part of SatsPath × Arkade MVP integration. See `satspath/README.md` for protocol overview.*